

STAT 5700 — Quiz 5**Date:** March 24, 2026**Name:** _____**Problem 1 (3pts)** Let $m(t) = \frac{1}{4}e^t + \frac{1}{4}e^{3t} + \frac{1}{2}e^{5t}$. Find the following:

- $E(Y)$
- $V(Y)$
- The probability distribution of Y

$$m'(t) = \frac{1}{4}e^t + \frac{3}{4}e^{3t} + \frac{5}{2}e^{5t}$$
$$E(Y) = m'(0) = \frac{1}{4} + \frac{3}{4} + \frac{5}{2} = \frac{7}{2}$$

$$m''(t) = \frac{1}{4}e^t + \frac{9}{4}e^{3t} + \frac{25}{2}e^{5t}$$
$$E(Y^2) = m''(0) = \frac{1}{4} + \frac{9}{4} + \frac{25}{2} = \frac{60}{4} = 15$$
$$V(Y) = 15 - \left(\frac{7}{2}\right)^2 = \frac{11}{4}$$

To find probability distribution of Y , recall that $m(t) = E(e^{ty}) = \sum_{y \in S} e^{ty}p(y)$

Since $m(t) = \frac{1}{4}e^t + \frac{1}{4}e^{3t} + \frac{1}{2}e^{5t}$, we can discern that the support of Y is 1,3,5, and $p(1) = 1/4$, $p(3) = 1/4$ and $p(5) = 1/2$

Y	$p(y)$
1	1/4
3	1/4
5	1/2

Problem 2 (2pts) Consider a new random variable with the following cdf

$$F(y) = \begin{cases} 0, & y < 0, \\ 1 - e^{-y}, & y \geq 0 \end{cases}$$

- a. Find $f(y)$

$$f(y) = F'(y) = e^{-y}, \quad y \geq 0$$

b. Find the 90th percentile of Y

$$F(Y) = 0.9 = 1 - e^{-y}$$

$$0.1 = e^{-y}$$

$$-\ln(0.1) = y$$

$$y = 2.302585$$